ActInf Textbook Group ~ Cohort 4 ~ Session 3 (Chapter 2 and 3) ~ 9/11/2025

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All right, welcome back everyone. It is September 11th, 2025 and we're in cohort 9 having a dual discussion talking about both the roads in chapter 2 and three. So, does anyone want to begin with a a section from chapter 2 or three that they want to go to or a question that they added to the questions table or a question that they saw in the table or just any other question or or topic related to low and the high Daniel. >> Yep. Go for it, Carrie. >> Um, there was a mention of reinforcement learning as a special case um in specifically in relationship to the low road and on page 37. Um, would you be willing to just sort of uh and and is that something that came out after active inference was sort of described as it or was that an intended part of the methodology to kind of explain or yeah explain the phenomena of reinforcement learning? >> Yeah, that's a that's a great question. I think I I am not exactly sure on the the the timing of the intentions, but that would be cool to to look at and review. Um we can see this expected free energy objective function as reconciling or integrating these two otherwise independent lines of model fitting. So in regular valueoriented decision theory or expected utility or reinforcement anything where there's a score and implicitly or explicitly the better you do on a score the I mean the whole point of the score is like you either want lower score like golf or higher score like baseball. So everything where all the complexity of a situation maps onto a single value that's ranked that has long been um a a a feature of economic type models. And reinforcement learning is a um procedure for using scorebased reward to update values. learning based upon how well they perform. It's like we're going to, you know, the baseball players that lead to us having a higher score in the game, we're going to keep them in. We're going to tweak things that lead to higher scores being reached. Um, usually that's framed in terms of taking states of the world and mapping them on states of the world or observations, mapping them onto a proposed reward function and then you seek to maximize reward on that reward function. Here we're not going to propose an auxilary or accessory reward function. We're just going to have this C preference distribution and directly map to that. And then the second part is the epistemic value which is that other lineage of of models that we're doing things like information maximizing or novelty based search. So it'd be interesting to see what um years or what intentions people came up with different things. Um, but big picture, yes, reinforcement learning, anything that's score-based is understood in this decision-making setting as aligning observations with preferences. And then there's this epistemic element which is not captured by preference orientation, but it's it's captured by learning or by information gain. Let's see what reinforcement learning. Here's that idea of of using a value function to rank or order. And there's a there's many flavors of this, but they all come down to doing some update procedure so that you get a higher score. Here is that move that they say is apparently strange but powerful, which is to forego this auxilary reward distribution. Directly connect observations to expectations. Now we're four, but restating that priors over observations do the work of a value or cost function. Anyone else? Just any we can go to a different question or any other idea. Um there's a question in the questions table uh that connects um the idea of persistent living systems um with uh economics or like financial um yeah I think it's the one with the orange text and it seems like then active inference is kind of a tutology like these systems exist because they minimize free energy and they minimize free energy in order to exist. So how how um yeah to think about this in a biological system versus a man-made engineered kind of approach to the system. Okay. Okay. So, from the book. Okay. Anyone want to add another just idea or aspect to this? to quote saying things living systems or persistent systems at whatever time scale have to be within some set of preferred states or just some state of characteristic some set of characteristic states. It's like if the thing weren't what it were it would either be a different thing or nothing at all. So this is the kind of tutology but that takes on a homeostatic flavor if we're talking about life. So then the question one can argue social systems have preferred states created by from both evolutionary niche and learn cognitive mechanisms. So just to the narrow question has active inference been applied to these sorts of systems. Yes, these are just a few of the of the references I guess that people have wrote in. they're at different stages of implementation in in terms of people developing um software models. But I mean when for example in in um like here's here's a a paper with a simulation that's based upon kind of like a gossip type model. Here's a paper that we um did in 2022 where we didn't provide any simulation but we just made it a um we basically did the chapter 6 thing which is we talked about different entity types defined them in terms of their their roles their affordances and and what kinds of situations they're engaged in. So it's a kind of it's like walking through the chapter six strategy like what are the kinds of things in the ecosystem what are their perception cognition and action affordances and then that kind of relates to to um some of the development work that we do to make flexible systems where we can go from having a specification of a complex multi- aent system to having executable simulation. So for some people um active inference hasn't been applied until the simulation is run and they have it in hand and so there there uh there's some limited work but at the level of just talking about scripts and social sciences then yes it it it very much has been applied. And there was I'll add the resource here for the um the social sciences course that we did in 2023. So here there were six uh six lecturers Ael, Ben, myself, Lorenna, Mal. Okay, I guess five and uh we each had a lecture and then a discussion section. So that was just covering some social topics. But this is this on 42. This is very much like the 0ero to2 jump where it's like how can a tautology uh be be useful? How can saying things are what they are be a useful um contribution? And that's where it's like that's that's where it goes from free energy principle explains nothing to it explains everything and then you know sometimes back into the zone of disillusionment and then where we march out to to utility and learning is just it's the central obvious but mysterious piece Steph or or Terry or Nick just continue on this or just any other section of two or three or question you want to explore. Daniel, I'll just uh offer a little bit of context for for where I'm at. Um, I've been meeting up with Andrew, uh, just to get a little bit more like one-on-one, well, one-on-one perspective on some of the math. And so, I've been bouncing around on a lot of stuff and I have a very like, I guess, specific set of questions. So, I've popped in today just to just to get some of the background like what's being talked about and see if things are landing. So, but I don't have any um I don't I I don't think I have any specific questions at the moment. Yeah. What parts of math have been fun? Sorry, I I didn't hear what she said. >> What parts of the map have been fun or or useful to learn? >> Oh, yeah. Like this is this has been I'm having a lot of fun. Uh so much application here and um getting getting um the the term free energy. I've just always started from like a very physical free energy perspective. So um I get what free energy means. It's just been um somewhat of a different domain to to think from from the beginning. Yeah, I think this this um this this was from 2021 when we discussed this paper of Alex Keeper. This is still probably one of the strongest uh places that's like no the free energy is the physical free energy. Let's see how like systems implementing variational inference is the downstream consequence arising from necess necessity or sufficiency of thermodynamic minimization. That that that claim is out there. And then but but that's what's so fascinating is it's obvious you can think of situations where um like if we're just talking about the frog jumping out of the hand in chapter that doesn't need to we don't need to look at how many calories the frog is burning. So it it's obvious that we can do approximate basian inference with a variational free energy for fantastical surreal pedagogical situations. >> can propose variational free energy functionals that do not have connection to real thermodynamic systems. So that that's an important starting position. It's kind of like squares and rectangles. So there are variational free energies that don't even need there's that they don't live or die by being tied to calories. >> Yes. >> The question is are there calorie consuming systems that what can we what then what can we say about calorie consuming systems? And that's I do want to I do want to say yeah that's what that is a central location for me and so I've I've gotten to Jane's paper on uh from 1957 [Music] um basically doing the maximum entropy principle. So um to establish where the concept jumps from statistical mechanics meaning physical thermodynamic entropy to Shannon information entropy the concepts are equivalent and one doesn't need to remain in the physical thermodynamic sense of the understanding and one can move simply into the information setting which is then where I feel justified to I I know where that location jump occurred and that's where I leave behind at least in in thinking about the frog jumping out of the hand I don't need to worry about calories I can simply think in information concepts and that was where I was looking. Um >> because because one more thing was I am um I am focused on um blockchain consensus mechanisms specifically proof of work where we are extracting maximal information from a thermodynamic process. us uh where we're attempting to and so I am I do need to know the physical and the information location where where it's occurring or like why I can be justified in jumping from thermodynamic toformational. Yeah. Yeah. There's there's a lot of pieces there. I think one important aspect that that comes up in the statistical thermodynamics is when are you jumping from micro states to macro states. So if we have um you know hot and cold water molecules in a chamber or or two two um two gas molecules that are in separate chamber. So it's like and then we you know open the divider. So we get this reproducible convergence to equilibrium that has these smooth macro properties because of the micro scale being driven by local energetic gradients. But then we can look at the the macro scale convergence properties as abstracted and supervening upon the micro composed of the micro interactions that are energetic. But by the time we're talking about the binary decision of the so we we could investigate down to the ATP molecule getting cleaved for the the object jumping or not. But this model is a coarse graining of an energetic model. So it has a lot to do with layers of description of the system or just like central limit theorem. You know, if we're gonna if we're gonna if we have a little um I I think it's called Plinko. This was one of the this is like a uh let's see if this is it. Yeah. So like >> yeah early early central limit theorem of statistics >> heavily used this type of logic which is like even if you drop just one at a time one ball through this at a time you will recapitulate the Gaussian distribution on the bottom. So it's it's not even just that it's about interactions amongst particles. It can also be particles dropped in one at a time with small differences. But then if we're talking about the distributional properties here, so we're looking at just the mean and the variance of the Gaussian, we could talk about that using variational Gaussian. But then if if we were interested in the particulate behavior mechanistically giving rise to what gets composed into a Gaussian, then we'd be talking about the potential energy function and the kinetic energy and all of that of collisions. But it it would be kind of a scale/category misfit to look for collisions in the descriptive summary statistics of the Gaussian because the Gaussian's mean invariance or the the free energy of us modeling the Gaussian is not colliding with anything physical. But temperature might matter for the collisions of the ball and the pegs. So you got to see when when are you looking at these micro scale and then when are you looking at the the macro which course grains over the energetically based micro interactions. on the Plinko example, the um if you assume no knowledge of what's going to happen, but you know the Plinko board exists, then you have a you have information established. You you have established a prior belief on what could happen assuming the boundaries that this is the only model. This is the board that we we care to consider. And on any individual ball drop, we can say there is a range of values with a set of probabilities for where it should land. And so our expectation value um for one ball drop the the the most likely thing we should say is that it's going to app it's going to fall into the middle band. Um when that ball falls and it falls in the in the law uh the the the most the furthest outward band. We can't be misled by that being the uh we we need to know that um if we were to redo this result hundreds of times that that one result doesn't say anything to skew our internal model to lead us to an erroneous um belief. That's that's basically what a p value is, which is like if you if if you have if you observe something that's one in a million one in a million times, it's an expected amount of surprising things. So you shouldn't your base distribution. I mean, you can see this when like like let's just say when you're doing the two-factor login on a website and you look just at whether there's going to be a double, a triple or quadruple number. So, it's like you'll see that, you know, usually there won't be any doubles or triples. Sometimes there might be a double. A little bit more rare is a triple. So rare things happen when you're looking when you're drawing random numbers or just you look at a random number generator and just say okay I'm waiting till I see this threedigit combination. So that's why we do statistics and work with distributions. But there are situations where you're interested in the thresholds and there still could be something important and like also these these are all homogeneous balls. But let's just say that some of them um had a bias to one side or another. Then in the tales you'd see massive enrichment. >> You know here out of out of 1200 one went all the way here or one sequence of coin flips led to a certain outcome. >> Yes. But >> that's where you >> seeing one outcome of five standard deviations over the appropriate number definitely should not update your distributional beliefs. >> Yeah. Or it's like one water you could pour water pour a cup of water into the into one side of the ocean and it's like there's going to be one molecule that seems like it takes a really you know amazing path to another location. But that's that's al that's called the Texas sharpshooter fallacy. which is basically you shoot and then you draw the bold side or that's also called like hypothesizing after results are known park those are so these are just sort of general statistical topics and they show how you know how you know there's different aspects of statistical epistemic hygiene which is where these models kind of take us to explore and basian statistics lets us be explicit about our prior beliefs. Whereas in frequentist statistics, you just take the observations plainly and you do descriptive modeling on what you observe. Let's just highspeed look over two and three and then anyone can can ask another dollar question. Okay. Low road first perception as inference. How can we think about just sensemaking and the incoming perception of the world in an inferential cognitive computational setting? Background on probability. All probabilities have to sum to one. Whether we're talking about perception of whether it's raining or not or later when we get to action, probabilities sum to one. They get reormalized to one. Example, unknown object jumping or not. It could be a frog or an apple. Works through this with exact phase. This is a situation where the numbers are small and the state spaces are small. So you can actually just exactly calculate that. However, you also might want to calculate some other features. Not just your posterior belief but how surprising and the two notions of surprise which are going gone into here which is how surprise how surprising is the observation which is kind of like what is the zcore or how many sigma is the observation is it is the ball that next observation ball falling right in the center of the distribution it's minimally surprising or how far is it falling out and then the second notion of surprise is how much do we update our beliefs following observation. Then still staying on this perception theme, they go to biological inference. Talk about what what's optimal about biological systems? What's optimal about biological inference? That brings us to this richer setting where we have the the particular partition getting brought up in terms of generative model, generative process. Then they go into action as inference. Okay, we looked at how we could use exact or approximate bays for perception. Turns out you can also do that for action inference. Minimizing the discrepancy between the model and the world can be done through this joint optimization of perceptual updating and action updating where you can't optimize it explicitly or act or or exactly. There are approximative techniques broadly. There's sampling based techniques and variational based techniques. If you're going to be making uh hybrid synthetic intelligence architectures, you might use any or all of the above. It's not about one being the only way to do it. It's just that here are some of the features of variational inference optimization. Here is that free energy variational free energy quantity that we've been discussing. not invented by the book. The math is elsewhere and is very standard in many statistical textbooks. It's great conversations to have with AI and people who like statistics. here is reframing that agent's world partition in terms of the agents beliefs Q and the observation Y coming in going into the free energy meat grinder and then the two ways to reduce that discrepancy flowing out of that changing the mind changing the world that's in the variational free energy setting which is kind of like real time perception oriented or instantaneous reflexive actionoriented expected free energy is an extension that moves into planning and explicit counterfactuals as imperence. It's a relevant feature if you want to make a model that does have explicit counterfactuals and look ahead and policy rollouts, but not all models have those. That's where we get to what we looked at earlier, which is looking at policy inference in terms of its pragmatic and epistemic components. And that's the end of the low three is going to pick up from a different position. It's going to converge on the same position. the same chapter 4 same act of inference which is like a free energy based energy based model generative model of a system cognitive system. So here the starting position is the partition or the boundary between the system and what surrounds it which you saw in chapter 2 but now this is going to play the central role which is okay. Now if you start from the partition and you think about the conditional dependencies or not and here the key the dog that doesn't bark in the night is that internal and external states are not connected directly. So there's no telekinesis and there's no telepathy. There isn't a direct connection except through the mediating blanket. And if there were those uh psychic phenomena, then they they could they could still be modeled by some other intermediating um you know node just just not to rule that out but but in this partition. And then there's two other important pieces that are not here. One of them is external states don't reach back to change action. So that embodies the assumption that action is under the control of the agent. And then internal states don't reach back symmetrically to influence sense. So there isn't the opportunity just to achieve your preferences in observations just by changing the number on the scale or by changing the photons on the retina. 3.2 two shows how you can have this indirectformational relationship that can be interpreted within a surprise minimization setting which has a direct connection to Hamiltonian principle of least action table 3.1 lines up basically what we've been discussing the statistical mechanical ontology the basian information information theory ontology and how that relates to cognitive modeling variational free energy gets reapproached now from that more physics perspective and then expecting free energy little bit of more big picture on what active inference does from a theoretical perspective and that's the high road so So, there's a lot in these chapters. Hopefully that wasn't too too fast, but what's a section to look at again or or any other idea or question somebody wants to go to There's a lot of lot of transportation based questions that could be asked like what other train stops are on this line? Are there other roads to active imprint? Are there other roads or forks coming out of these nodes? Hey Daniel, could you give context for that image you just put up because I I um I've not seen that before. No, the that um the active interest in the center with the tree. >> This one? >> Yeah. Can you give me a little context for this? >> Yes. It's um figure 1.2. This is from chapter 1. And it's kind of the it's like the graphical abstract of the book in a way because it lays out, okay, here are the the high and low road and then here are some of the train stops along that way. Um, it it's not intended as as far as I read it to be exhaustive or to to be strict about the ordering exactly, but it does summarize how chapters two and three are going to unfold, which is starting on low road. Okay, we're doing something statistical. we're we're dealing with probabilistic systems or situations where there's partial information or just empirical settings. Oh, side note, I I I love this part because statistics can seem very theoretical and abstract, but it is the most real and grounded. Like statistics is how we deal with real empirical data. So if someone thinks they have another way to deal with real empirical data, then amazing. they're either contributing to statistics or they're making some other cousin field. But it's like this is how we deal with real systems is with statistics. So then we can think about different cognitive functions under this aspiration of having a unified cognitive modeling approach where we can deal with broadly the perceptual cognitive and action selection features know the inbound, the inside and the outbound features of systems. and all of the other accessories like attention, memory, anticipation, planning, like all those different features, deal with them within by thinking of it as a joint basian distribution. And then if you can calculate exact basian inference, great. But if you want to take this approximation that allows you to do incremental optimization, you want to use this common approximation method to that basian joint distribution called variational base. And then you reach active inference, which is okay, we have a joint basian distribution which we're updating according to the rules of variational inference. That's how we go from the grounded necessity to model or design real cognitive systems in terms of their different features as infrareds getting to this joint distribution and then if we want to approximate it we get the joint distributions that are the generative model of active imprints and then that's chapter two and then chapter three starts from what was there but not highlighted in that agent's niche partition, but it's going to say, "Okay, let's start from the idea of a partitioned graphical system and then interpret that boundary as being something that the system on one side is or really on either side is going to be doing evidencing across and then I would say self-organization, autopois niche construction do not really come up in chapter 3. But coming from the high road, we start from the partition and on the flows of information across the partition and then we get to literally the same location which is a joint distribution describing the agent in the environment which can be approximated with a variational free energy value. That's why there's And then that's chapter four. And I think this question Yes. Here's the paper where the high and low road uh were introduced 2019 beyond the desert landscape. And it's like this kind of this philosophical debate and they always have um fun you know terminology but I think somebody raised a critique well active inference is like a desert landscape. It's beautiful but it's austere. I think it might have been Andy Clark. But the agenda in active influence is to link formal descriptions of dynamical systems to a description of active perception in terms of beliefs and goals. that is the agenda and then here the the low road is described as Helm holds or contian. Now, personally, the last name as adjective I don't find massively informative because it can often refer to stances that the person didn't even take, but they alluded to or were later developed. So, it doesn't add information directly, but it gives a good pointer. And so, that was the high and low road, but it's closed source. Maybe someone uploaded it. And then here here's an example which is always useful to see which is like it can be terms that are are seemingly novel but they're all terms that were that that can be reused from the active inference ontology. So all the blue words are where we're using like this controlled vocabulary. And that also supports translation because we have translations for the ontology into a number of languages. And it supports formalizations as well because these these can be taken beyond being just a natural language expression and and brought into a more formal setting. High road seems more goal driven and strategic while low road seems more response driven and tactical. Yeah, as always there's a lot of fun comments in these. >> Um, can I uh connect this back to the Marov blanket comment you made earlier? You said that um action can't uh reach back in and change the ex the internal and inference can't reach and change the external states. Um but how does that connect to the high and low road and where does the marov blanket kind of appear in the concept? Yeah, that's a very deep question. Um, it comes down to how the modeler wants to topologically partition the agent in the niche. the thing from what is not the thing. How do you want to partition it by talking about the thing at all? You have already identified that there's some kind of differentiating discreetly differing components. That's what is going to be described with the connectivity of of the topology. So then how many nodes are there going to be? And if we're going to see it as a directed graph, which arrow heads are we going to allow? you could allow all of the edges. It's still it still could be done. But if you ran that computational model, you would get some outcomes that might not be that useful or interpretable. Like if we were going to have here's us in the room and then there's the temperature of the room and then we're going to get the temperature observation and turn on the heater or the cooler. So if we allow your belief about the room just to in this sort of um telekinetic effect just change the temperature like the statistical model will just do that. So if that's the situation you're describing, that is fine. But um this is this is not the only topology of how this situation can be analyzed. It just embodies like these kind of two core things. Again, the first one being that we're we're talking about systems that don't that that do have a mediation step, even if it's just kind of a pass through. And then um you are at least in control of dispatching actions. There can be some situations where the actions efficacy stying but at least the request is made under your control. And then the thumb on the scale is the backwards perception which is if you can just take a deceptive if if you allow you know then you could just change what you're seeing without changing the situation which basically is a short-term win but then it breaks the the validity of the relationship between the temperature of the room and the thermometer. So that you just broke your thermometer to control the numbers that it's showing. So So what's missing here embody those kind of common sense type constraints, but they're not actually required. And you could have a low road or a high road reaching a graphical model that has a different topology. So this isn't the only nodes or arrows that reflect models. Like if we look at the figures here, these are other graphical models that reflect other causal models of the world. So those could be constructed via the low road and those could be under the free energy principle of the high road. So other topologies are required or are possible but this one is kind of a classic but in practice like these edges are often not even engaged. It's sort of like well okay there's a edge between the thermometer and the air conditioner that doesn't go through the decision maker. So this is kind of this ongoing discussion which is how what how do we things from their environments in a way that's flexible? But answer is you can make whatever graph you want. But the graph should embody some of the constraints that you're looking to operate under. And you can make graphs that wildly diverge from realistic situations, which makes them less useful, but but doesn't make them invalid. Daniel, does this have anything to do with the nature of Markoff um blankets and chains in terms of the like in my mind I'm thinking like we can't time travel like you can't go back in time and alter circumstances. And so that's part of why we also don't have an impact. I mean that I guess my question is which which comes first? The the way that we're setting up the the boundaries and therefore assigning Markoff chains or or are we expecting or making an assumption or setting up as the baseline that they do adhere something to a Markoff blanket and therefore this is do you see what I'm saying? >> Yeah. Yes. And then there's this like reification question like well what if we are modeling time in a way that constrains us and so that we can't do time travel but otherwise we could do time travel or something like that like um yes a lot of ways to take it but you know if if we're modeling the the uh a Gaussian distribution the only two we have are where is it located and how wide. So if the distribution were like three humps, we could do our best, but we're never going to break out of the galaxy into the three humps unless we made a structure learning model. It just so um so similarly a model will not tell you that it is inadequate. It you it will just allow itself to be used in its mo in its best possible way. So yes. So then then you go okay so now we're going to deal with models of time and we're going to treat time as a markoff chain which is to say the past only influences the future through the present. So it's like past present and future. Now there can be a very thick present that brings a lot through like in the extreme case the whole thing gets brought through or in the minimal case only a summary is brought through. Um, so then then the question is like is that just a convenience? Because if we're going to do a 100 time steps of a simulation, certainly it's more convenient to just do 99 moments versus if each moment can influence each other moment, then there's going to be vastly more calculations. So it's more convenient to have the markoff property and it seems to accord with how we think causality actually flows but you don't exactly know. But the cool thing is you could test it and you could say okay I'm going to propose an alternate topology for time and see if that provides an explanation or a prediction that I didn't get otherwise. And then if you have evidence for that, then there's evidence for that. Okay. Hope that was uh useful. At this time next week, it will be chapter 8 on continuous models. And uh see you all then. >> Thanks Daniel. This is awesome. >> Bye.